

DATA ANALYTICS

PYTHON ASSIGNMENT 2 :DATA STRUCTURES- LIST, DICTIONARY, SET & CONDITIONAL STATEMENTS

SUBMISSION:03.10.2026

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Data Structures - List, Dictionary, Set & Conditional Statements

List (Creation, Modification and Access):

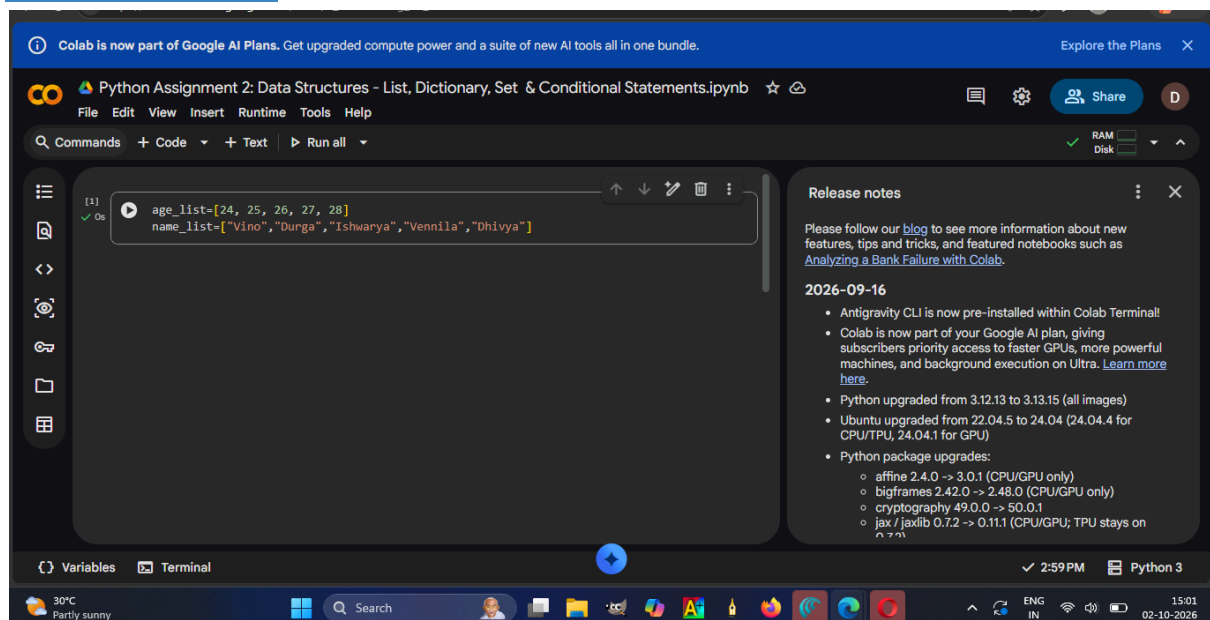
1. List Creation:

Task:

- Create a list named `age_list` with five integer elements. For eg., [24, 25, 26, 27, 28]
- Create a list named `name_list` with five string elements.

Solution :

Preview screenshot :



Explanation:

- First I had created the list with five integer elements and named the list as `age_list`.

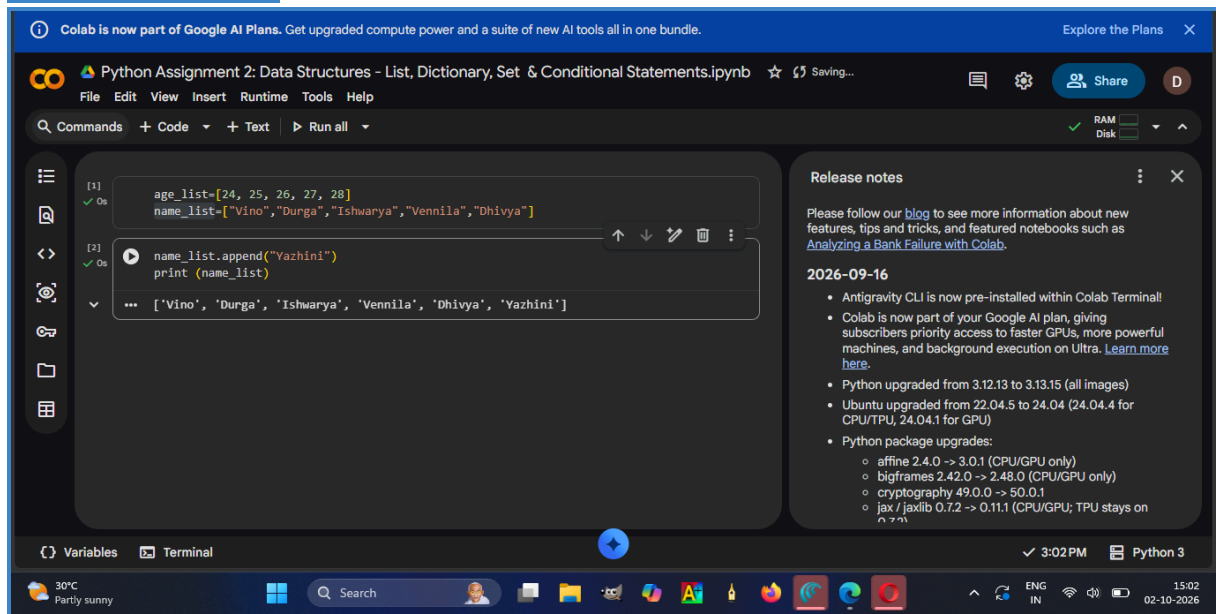
- Then created the name_list with five string elements.
- The above mentioned screenshot represents the result.

2. List Operations / Modifications:

Task 1: Append the string "Yazhini" to name_list.

Solution :

Preview screenshot :



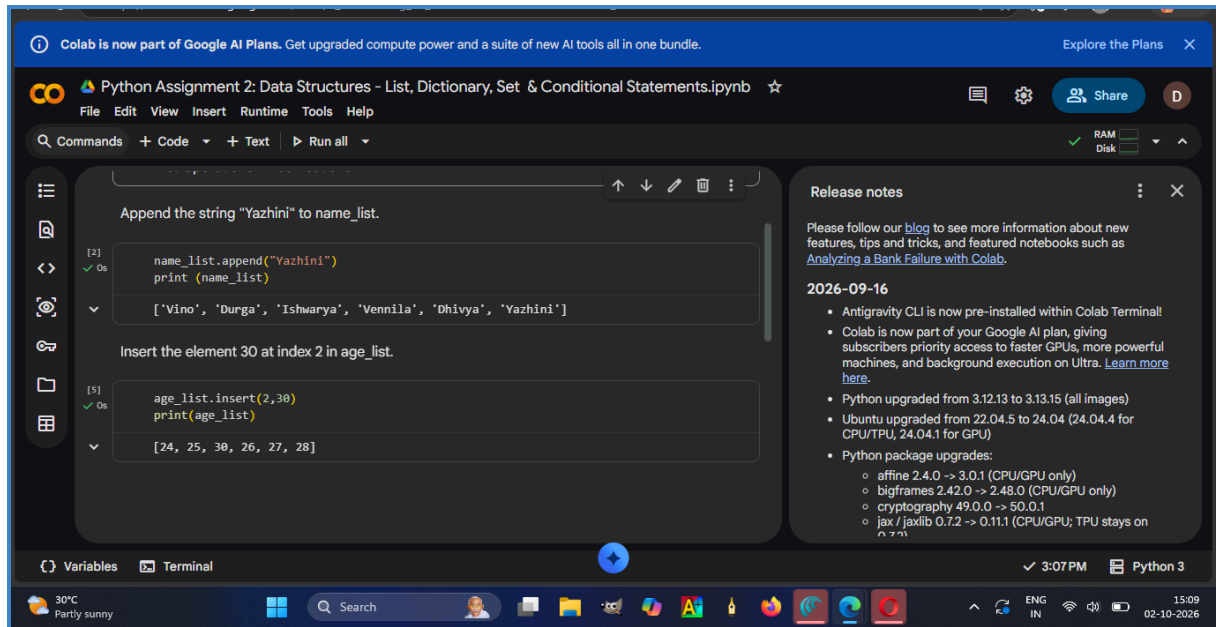
Explanation :

- I had used the append method for adding a string to name_list.
- The above mentioned screenshot represents the result.

Task 2 : Insert the element 30 at index 2 in age_list.

Solution:

Preview screenshot :



The screenshot shows a Google Colab notebook titled "Python Assignment 2: Data Structures - List, Dictionary, Set & Conditional Statements.ipynb". The notebook has two code cells. The first cell, labeled [2], contains the code to append the string "Yazhini" to a list named name_list and print it. The output shows the list ['Vino', 'Durga', 'Ishwarya', 'Vennila', 'Dhivya', 'Yazhini']. The second cell, labeled [5], contains the code to insert the element 30 at index 2 of a list named age_list and print it. The output shows the list [24, 25, 30, 26, 27, 28]. On the right side of the notebook, there is a "Release notes" panel for the date 2026-09-16, listing updates to Colab, Python, and various packages. The bottom of the screen shows a Windows taskbar with the date 02-10-2026 and time 15:09.

```
Append the string "Yazhini" to name_list.  
[2]  
✓ 0s  
name_list.append("Yazhini")  
print (name_list)  
['Vino', 'Durga', 'Ishwarya', 'Vennila', 'Dhivya', 'Yazhini']  
  
Insert the element 30 at index 2 in age_list.  
[5]  
✓ 0s  
age_list.insert(2,30)  
print(age_list)  
[24, 25, 30, 26, 27, 28]
```

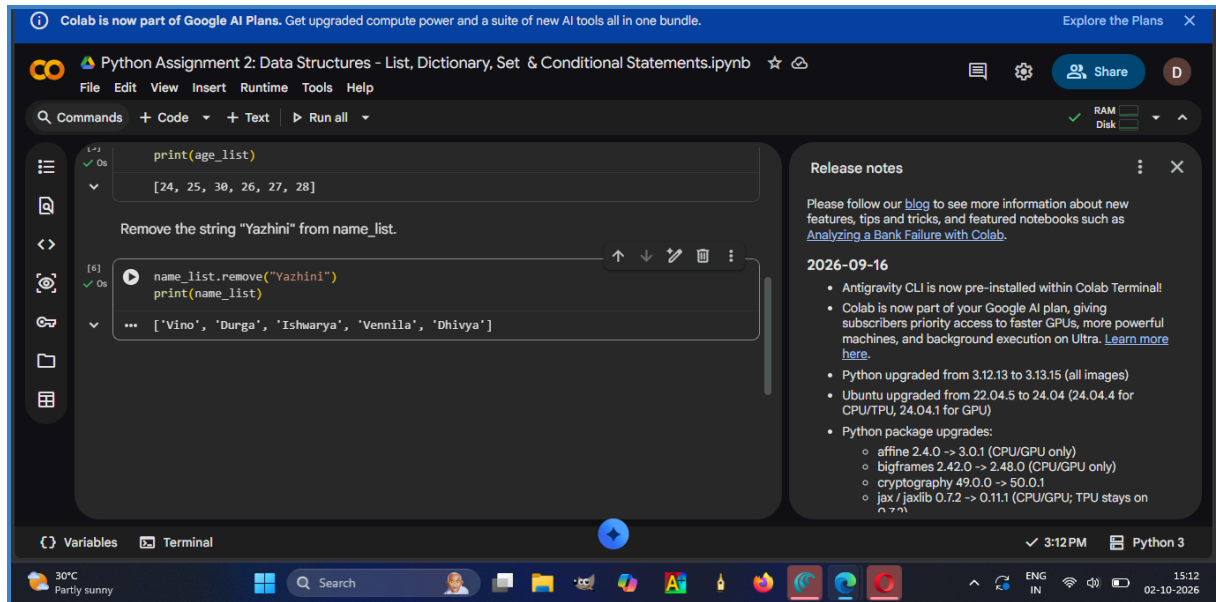
Explanation:

- I had used the insert method for inserting an element into age_list.
- The above mentioned screenshot represents the result.

Task 3: Remove the string "Yazhini" from name_list.

Solution:

Preview screenshot :



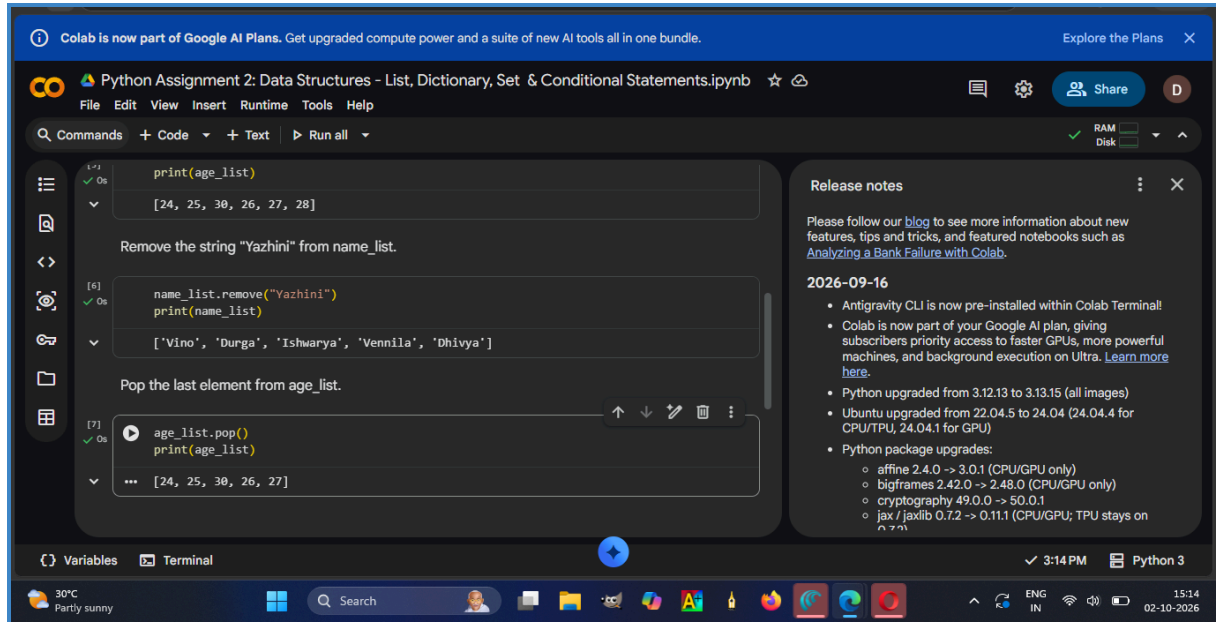
Explanation :

- I had used the insert method for inserting a string into name_list.
- The above mentioned screenshot represents the result.

Task 4: Pop the last element from age_list.

Solution:

Preview screenshot :



The screenshot shows a Google Colab notebook titled "Python Assignment 2: Data Structures - List, Dictionary, Set & Conditional Statements.ipynb". The notebook has three code cells. The first cell prints the initial age_list: [24, 25, 30, 26, 27, 28]. The second cell removes the string "Yazhini" from name_list and prints the updated list: ['Vino', 'Durga', 'Ishwarya', 'Vennila', 'Dhivya']. The third cell pops the last element from age_list and prints the updated list: [24, 25, 30, 26, 27]. A "Release notes" sidebar is visible on the right, dated 2026-09-16, listing updates to Colab, Python, and various packages.

```
[1] print(age_list)
[2] [24, 25, 30, 26, 27, 28]

Remove the string "Yazhini" from name_list.

[6] name_list.remove("Yazhini")
    print(name_list)
[7] ['Vino', 'Durga', 'Ishwarya', 'Vennila', 'Dhivya']

Pop the last element from age_list.

[7] age_list.pop()
    print(age_list)
[8] [24, 25, 30, 26, 27]
```

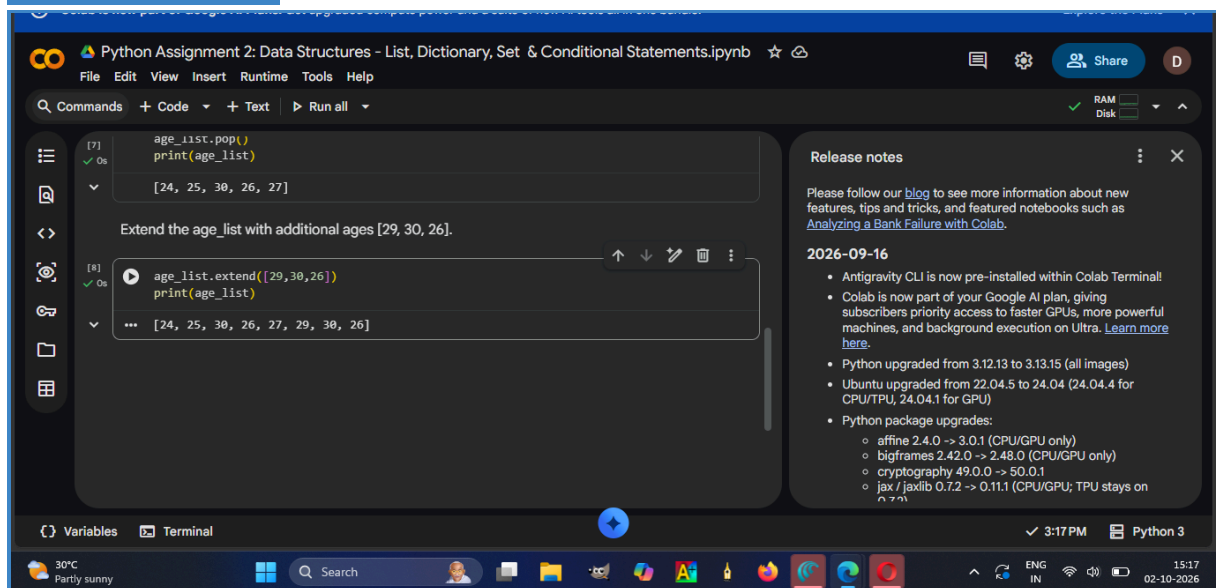
Explanation :

- I had used the pop method for deleting an element into age_list.
- The above mentioned screenshot represents the result.

Task 5 : Extend the age_list with additional ages [29, 30, 26].

Solution :

Preview screenshot :



The screenshot shows the same Google Colab notebook as before, but with a fourth code cell. This cell extends the age_list with the additional ages [29, 30, 26] and prints the updated list: [24, 25, 30, 26, 27, 29, 30, 26]. The "Release notes" sidebar remains on the right.

```
[7] age_list.pop()
    print(age_list)
[8] [24, 25, 30, 26, 27]

Extend the age_list with additional ages [29, 30, 26].

[8] age_list.extend([29,30,26])
    print(age_list)
[9] [24, 25, 30, 26, 27, 29, 30, 26]
```

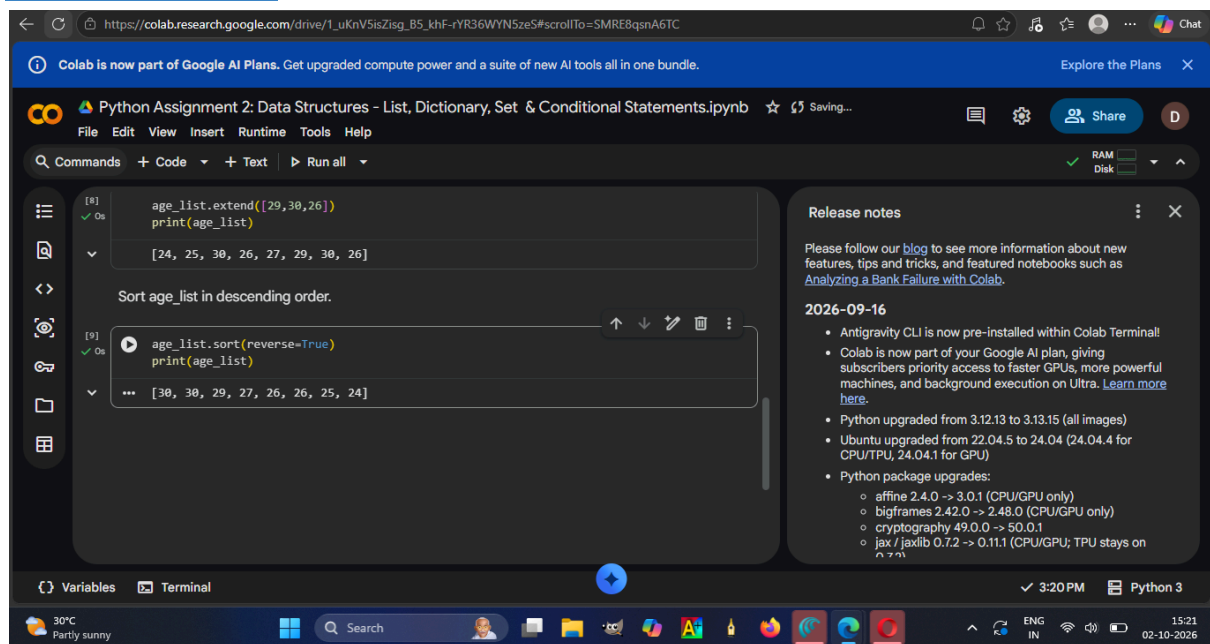
Explanation :

- I had used the extend method to add [29, 30, 26] into age_list.
- The above mentioned screenshot represents the result.

Task 6 : Sort age_list in descending order.

Solution :

Preview screenshot :



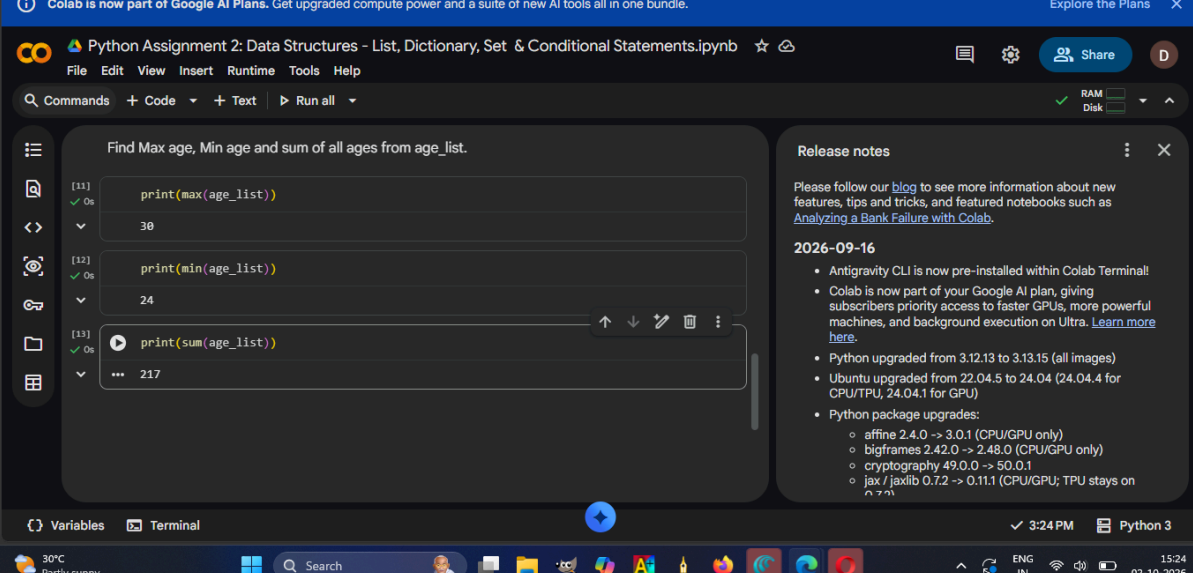
Explanation :

- I had used the sort method to sort age_list with reverse=true so it sorted the age list in descending order.
- The above mentioned screenshot represents the result.

Task 7: Find Max age, Min age and sum of all ages from age_list.

Solution :

Preview screenshot :



```
Python Assignment 2: Data Structures - List, Dictionary, Set & Conditional Statements.ipynb
```

```
[11]: print(max(age_list))
      30

[12]: print(min(age_list))
      24

[13]: print(sum(age_list))
      ... 217
```

Release notes

Please follow our [blog](#) to see more information about new features, tips and tricks, and featured notebooks such as [Analyzing a Bank Failure with Colab](#).

2026-09-16

- Antigravity CLI is now pre-installed within Colab Terminal!
- Colab is now part of your Google AI plan, giving subscribers priority access to faster GPUs, more powerful machines, and background execution on Ultra. [Learn more here.](#)
- Python upgraded from 3.12.13 to 3.13.15 (all images)
- Ubuntu upgraded from 22.04.5 to 24.04 (24.04.4 for CPU/TPU, 24.04.1 for GPU)
- Python package upgrades:
 - affine 2.4.0 -> 3.0.1 (CPU/GPU only)
 - bigframes 2.42.0 -> 2.48.0 (CPU/GPU only)
 - cryptography 49.0.0 -> 50.0.1
 - jax / jaxlib 0.7.2 -> 0.11.1 (CPU/GPU; TPU stays on 0.7.2)

Explanation :

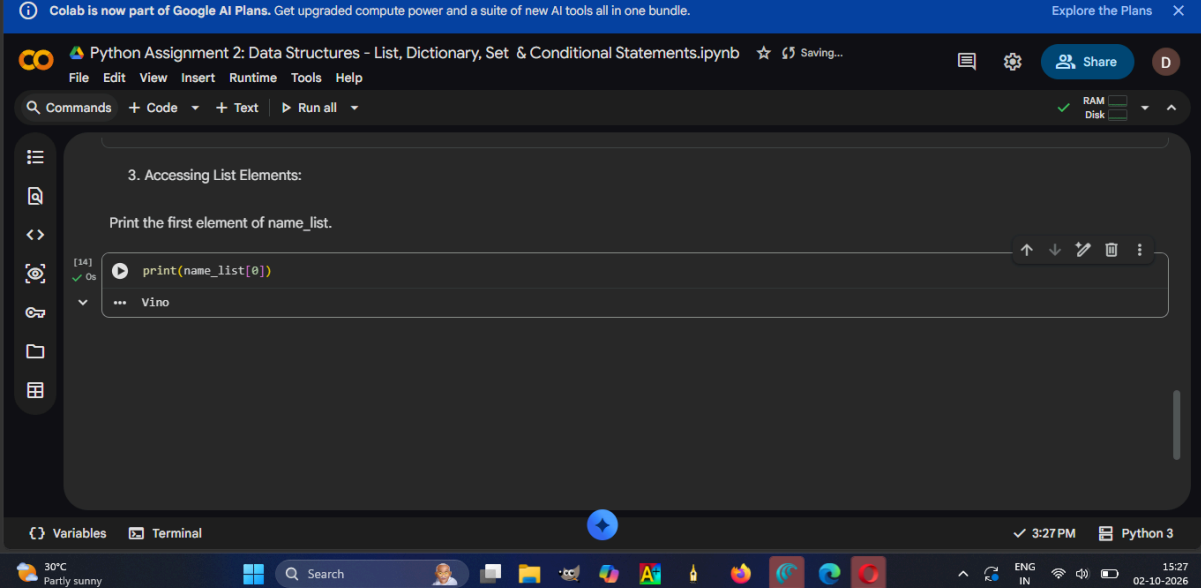
- I had used the max method to find the maximum age from the age_list. And a method to find minimum age from the age_list. And a sum method to find the sum of all ages from age_list.
- The above mentioned screenshot represents the result.

3. Accessing List Elements:

Task 1: Print the first element of name_list.

Solution :

Preview screenshot :



The screenshot shows a Google Colab notebook interface. The title bar indicates it's a Python Assignment 2 notebook. The code cell contains the instruction 'Print the first element of name_list.' and the code `print(name_list[0])`. The output shows the name 'Vino'. The interface includes a menu bar, a toolbar, and a status bar at the bottom.

```
3. Accessing List Elements:

Print the first element of name_list.

[14] print(name_list[0])
✓ Os
... Vino
```

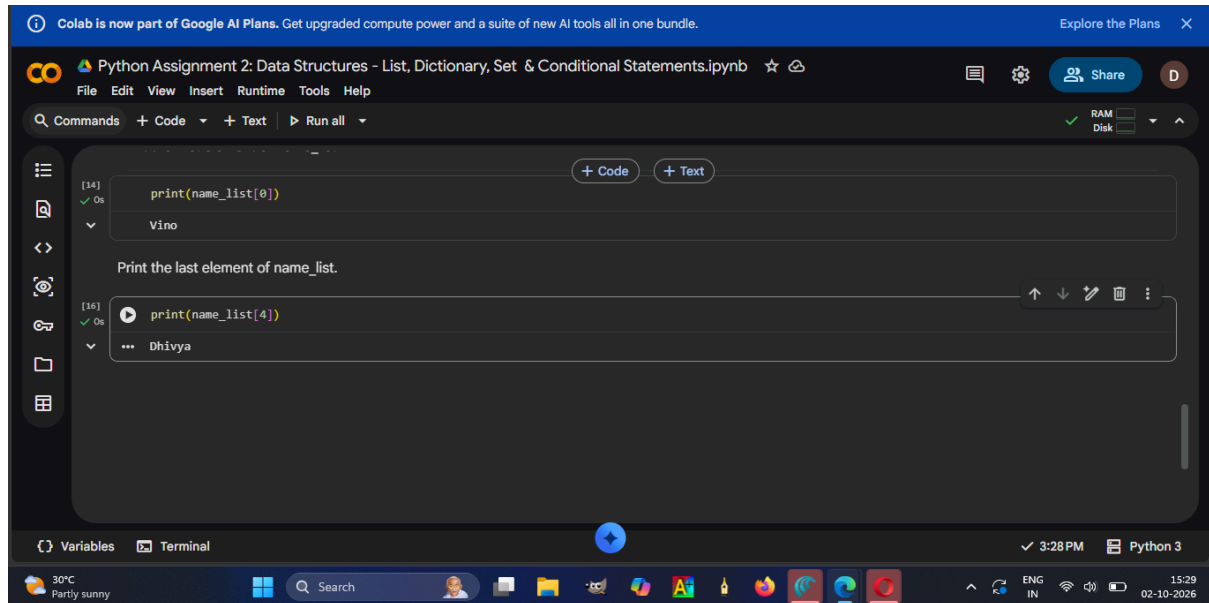
Explanation :

- I had used the indexing method for printing the first element in name_list.
- Here 0 represents the first element in the name_list.
- The above mentioned screenshot represents the result.

Task 2:Print the last element of name_list.

Solution :

Preview screenshot :



The screenshot shows a Google Colab notebook titled "Python Assignment 2: Data Structures - List, Dictionary, Set & Conditional Statements.ipynb". It contains two code cells. The first cell, labeled [14], contains the code `print(name_list[0])` and has executed successfully, outputting "Vino". The second cell, labeled [16], contains the comment "Print the last element of name_list." followed by the code `print(name_list[4])`. This cell has also executed successfully, outputting "Dhivya". The notebook interface includes a menu bar (File, Edit, View, Insert, Runtime, Tools, Help), a toolbar with icons for file operations and execution, and a status bar at the bottom showing system information like temperature (30°C) and time (15:29).

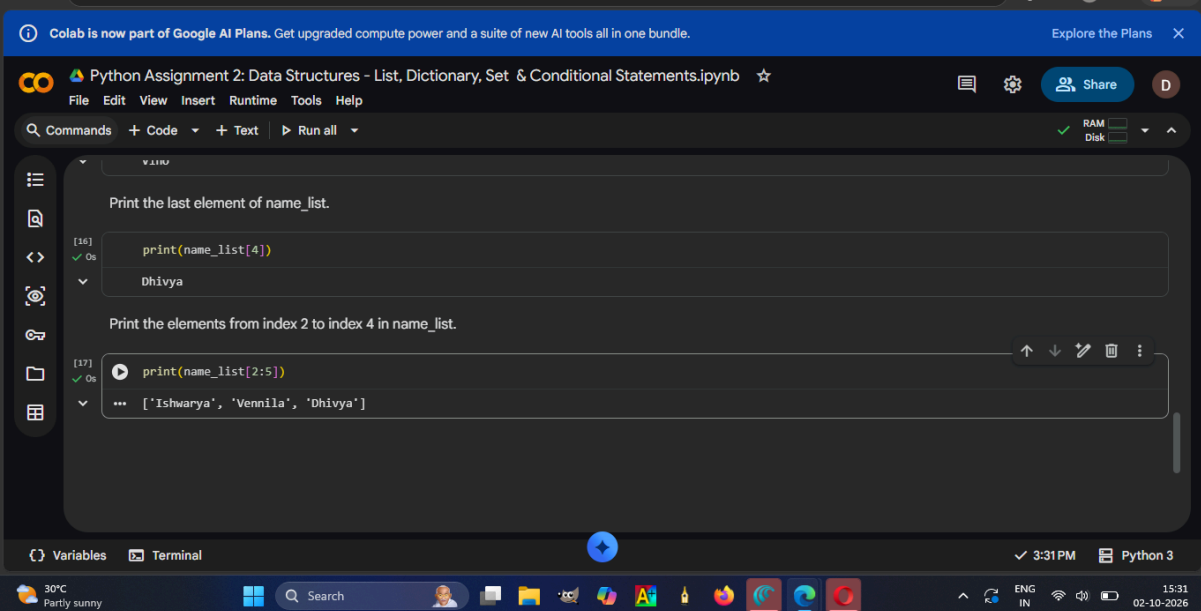
Explanation :

- I had used the indexing method for printing the last element in name_list.
- Here 4 represents the last element in the name_list.
- The above mentioned screenshot represents the result.

Task 3 :Print the elements from index 2 to index 4 in name_list.

Solution :

Preview screenshot :



The screenshot shows a Google Colab notebook titled "Python Assignment 2: Data Structures - List, Dictionary, Set & Conditional Statements.ipynb". It contains two code cells. The first cell, labeled [16], has the comment "Print the last element of name_list." and the code `print(name_list[4])`, which outputs "Dhivya". The second cell, labeled [17], has the comment "Print the elements from index 2 to index 4 in name_list." and the code `print(name_list[2:5])`, which outputs the list `['Ishwarya', 'Vennila', 'Dhivya']`. The notebook interface includes a menu bar (File, Edit, View, Insert, Runtime, Tools, Help), a toolbar with icons for file operations and execution, and a status bar at the bottom showing system information and the current time (3:31 PM).

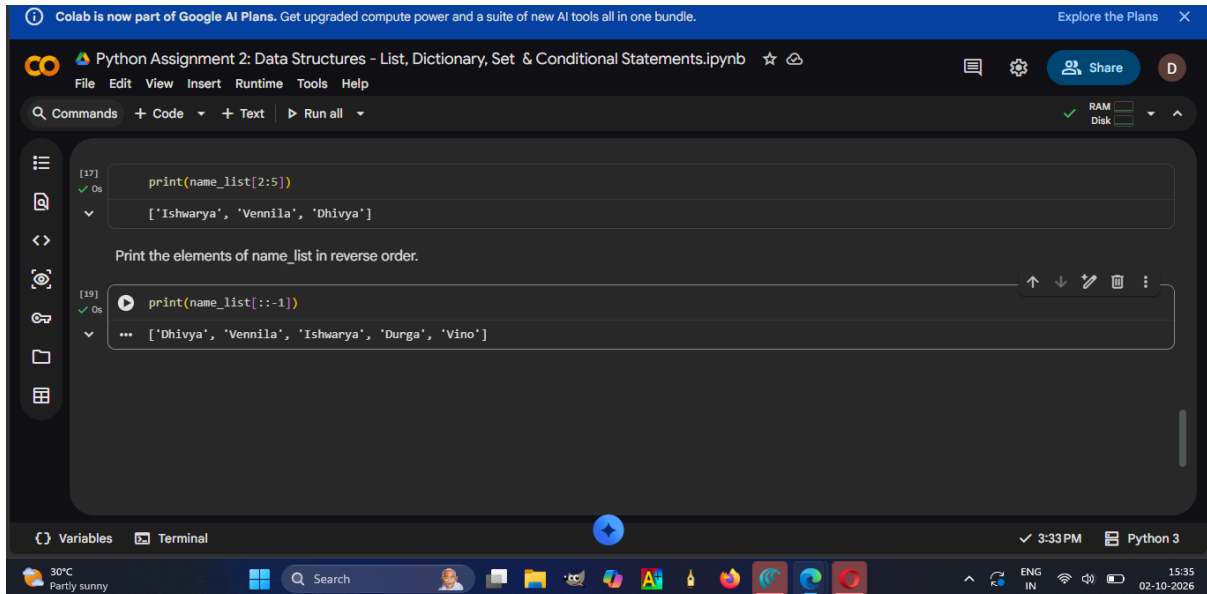
Explanation :

- I had used the slicing method for printing the 2 to 4th element in name_list.
- Here 2 represents the third element 5 represents the last element in the name_list.
- The above mentioned screenshot represents the result.

Task 4: Print the elements of name_list in reverse order.

Solution :

Preview screenshot :



The screenshot shows a Google Colab notebook titled "Python Assignment 2: Data Structures - List, Dictionary, Set & Conditional Statements.ipynb". The notebook has two code cells. The first cell, at index 17, contains the code `print(name_list[2:5])` and its output is `['Ishwarya', 'Vennila', 'Dhivya']`. The second cell, at index 19, contains the code `print(name_list[::-1])` and its output is `['Dhivya', 'Vennila', 'Ishwarya', 'Durga', 'Vino']`. The text "Print the elements of name_list in reverse order." is written above the second code cell. The notebook interface includes a menu bar (File, Edit, View, Insert, Runtime, Tools, Help), a toolbar with icons for running and saving, and a status bar at the bottom showing the time as 3:33 PM and the Python version as Python 3. The Windows taskbar is visible at the very bottom of the image.

Explanation :

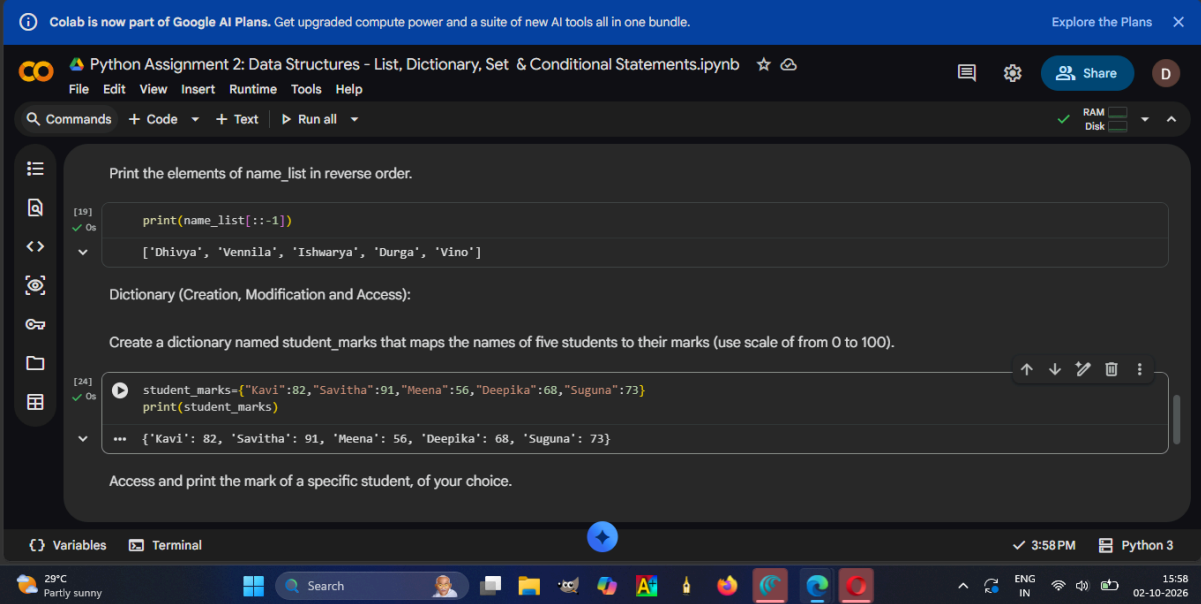
- I had used the slicing method for printing the name_list in reverse order.
- Here `::-1` is used to reverse the name_list.
- The above mentioned screenshot represents the result.

Dictionary (Creation, Modification and Access):

Task 1 : Create a dictionary named `student_marks` that maps the names of five students to their marks (use scale of from 0 to 100).

Solution :

Preview screenshot :



The screenshot shows a Google Colab notebook interface. The title bar indicates it's part of Google AI Plans. The notebook is titled "Python Assignment 2: Data Structures - List, Dictionary, Set & Conditional Statements.ipynb". The code is as follows:

```
[19] print(name_list[::-1])
[19] ✓ 0s
['Dhivya', 'Vennila', 'Ishwarya', 'Durga', 'Vino']

Dictionary (Creation, Modification and Access):

Create a dictionary named student_marks that maps the names of five students to their marks (use scale of from 0 to 100).

[24] student_marks={"Kavi":82,"Savitha":91,"Meena":56,"Deepika":68,"Suguna":73}
[24] ✓ 0s
print(student_marks)

*** {'Kavi': 82, 'Savitha': 91, 'Meena': 56, 'Deepika': 68, 'Suguna': 73}

Access and print the mark of a specific student, of your choice.
```

The bottom of the notebook shows the variable `student_marks` with its value: `{'Kavi': 82, 'Savitha': 91, 'Meena': 56, 'Deepika': 68, 'Suguna': 73}`. The status bar at the bottom indicates the system is at 29°C, partly sunny, and the time is 3:58 PM on 02-10-2026.

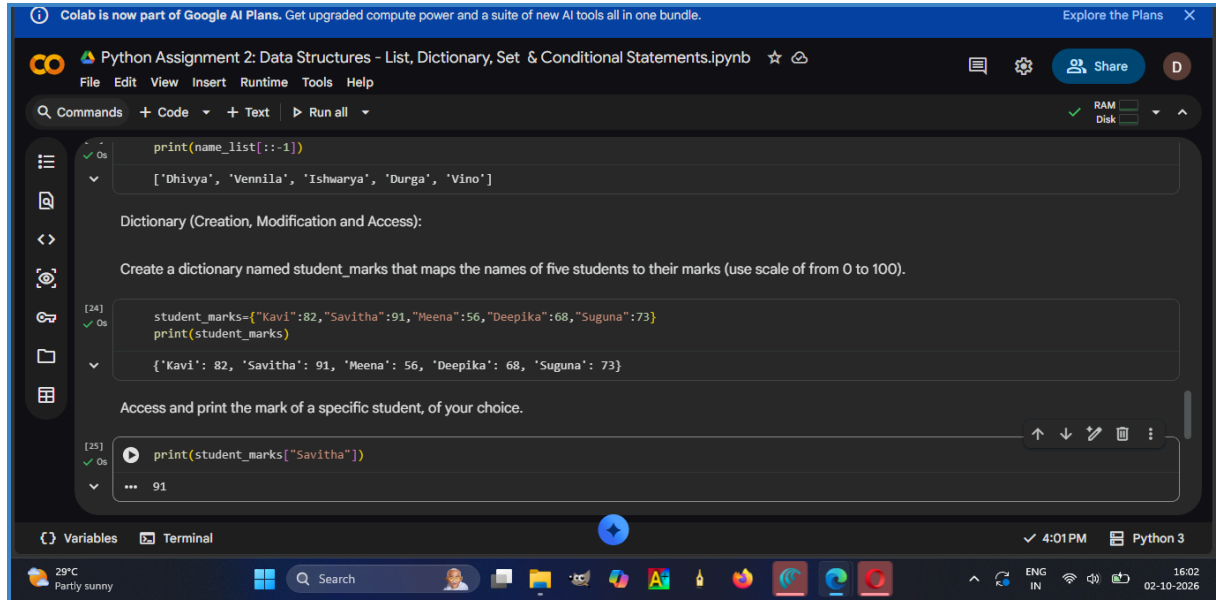
Explanation :

- I had created a dictionary named `student_marks`.
- Here key value pairs are separated by `' : '`.
- The above mentioned screenshot represents the result.

Task 2: Access and print the mark of a specific student, of your choice.

Solution :

Preview screenshot :



The screenshot shows a Google Colab notebook titled "Python Assignment 2: Data Structures - List, Dictionary, Set & Conditional Statements.ipynb". The notebook contains the following code and output:

```
print(name_list[::-1])
```

```
['Dhivya', 'Vennila', 'Ishwarya', 'Durga', 'Vino']
```

Dictionary (Creation, Modification and Access):

Create a dictionary named student_marks that maps the names of five students to their marks (use scale of from 0 to 100).

```
student_marks={"Kavi":82,"Savitha":91,"Meena":56,"Deepika":68,"Suguna":73}
```

```
print(student_marks)
```

```
{'Kavi': 82, 'Savitha': 91, 'Meena': 56, 'Deepika': 68, 'Suguna': 73}
```

Access and print the mark of a specific student, of your choice.

```
print(student_marks["Savitha"])
```

```
91
```

The bottom of the screenshot shows the Windows taskbar with the date 02-10-2026 and time 16:02.

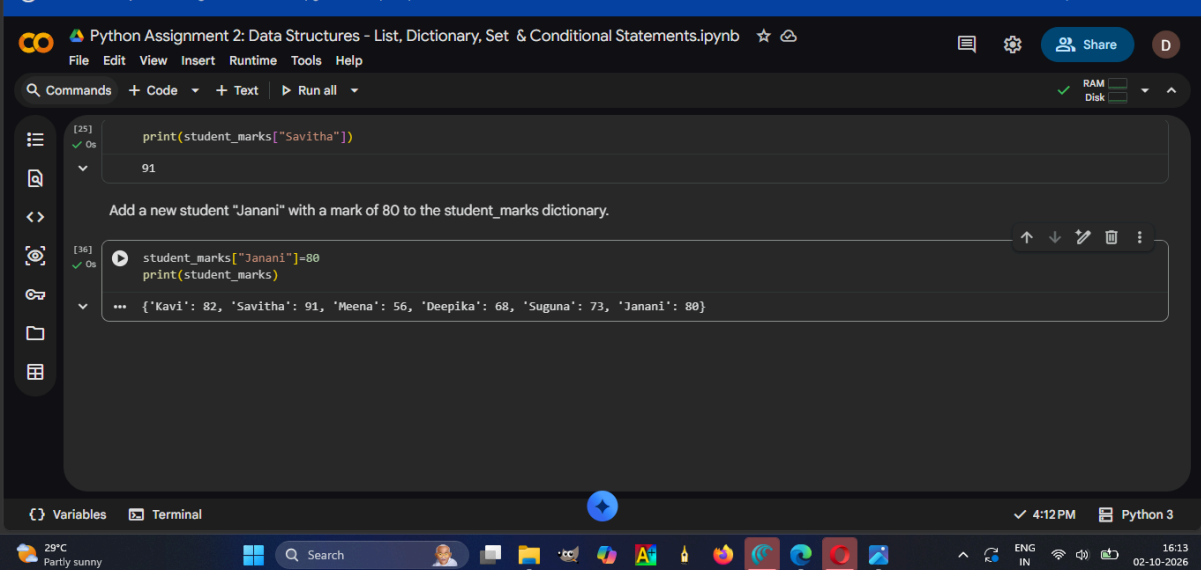
Explanation :

- I had used the key ("Savitha") to print the value (91) in student_marks.
- Here the student name is key and the mark is a value.
- The above mentioned screenshot represents the result.

Task 3 : Add a new student "Janani" with a mark of 80 to the student_marks dictionary.

Solution :

Preview screenshot :



The screenshot shows a Jupyter Notebook interface with the title "Python Assignment 2: Data Structures - List, Dictionary, Set & Conditional Statements.ipynb". The notebook has two code cells. The first cell contains the code `print(student_marks["Savitha"])` and its output is `91`. The second cell contains the code `student_marks["Janani"] = 80` followed by `print(student_marks)`. Below the code, the output shows the updated dictionary: `{'Kavi': 82, 'Savitha': 91, 'Meena': 56, 'Deepika': 68, 'Suguna': 73, 'Janani': 80}`. The interface includes a sidebar with icons for file management, a top bar with menu options (File, Edit, View, Insert, Runtime, Tools, Help) and a "Share" button, and a bottom bar with system information (29°C, Partly sunny, 4:12 PM, Python 3) and a taskbar with various application icons.

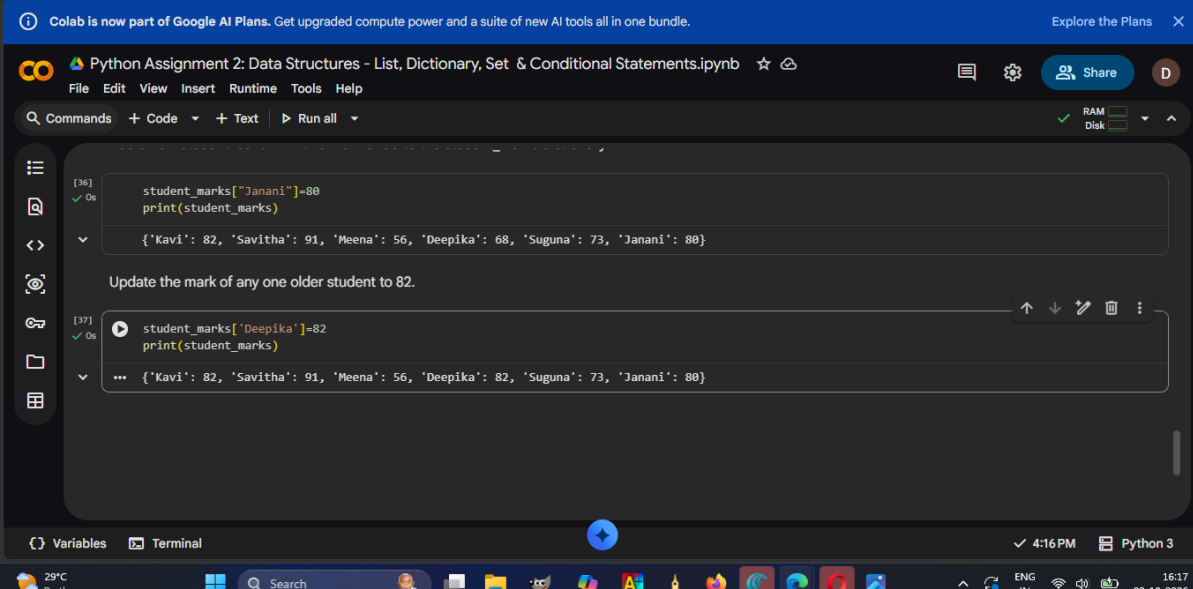
Explanation :

- I had added the student name "Janani" with mark 80 to student_marks.
- The above mentioned screenshot represents the result.

Task 4 : Update the mark of any one older student to 82.

Solution :

Preview screenshot :



The screenshot shows a Google Colab notebook titled "Python Assignment 2: Data Structures - List, Dictionary, Set & Conditional Statements.ipynb". The notebook has two code cells. The first cell, labeled [36], contains the code `student_marks["Janani"]=80` and `print(student_marks)`, which outputs the dictionary `{'Kavi': 82, 'Savitha': 91, 'Meena': 56, 'Deepika': 68, 'Suguna': 73, 'Janani': 80}`. The second cell, labeled [37], contains the text "Update the mark of any one older student to 82." followed by the code `student_marks['Deepika']=82` and `print(student_marks)`, which outputs the updated dictionary `{'Kavi': 82, 'Savitha': 91, 'Meena': 56, 'Deepika': 82, 'Suguna': 73, 'Janani': 80}`. The notebook interface includes a menu bar (File, Edit, View, Insert, Runtime, Tools, Help), a toolbar with icons for commands, code, text, and running all cells, and a status bar at the bottom showing the current time as 4:16 PM and the Python version as Python 3. The Windows taskbar is visible at the very bottom of the image.

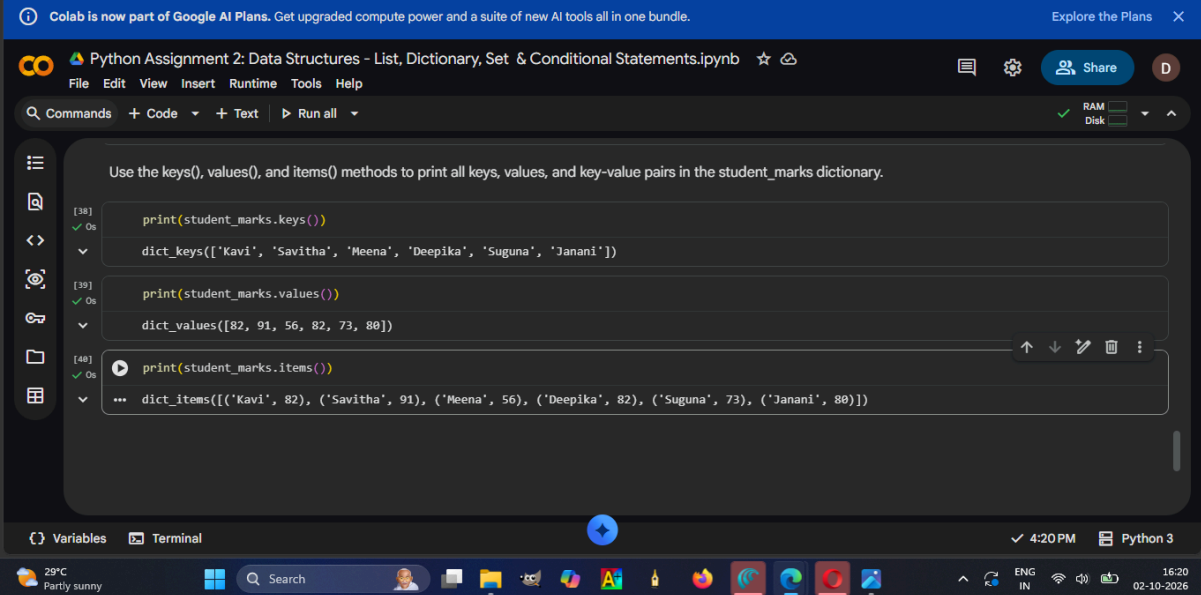
Explanation :

- I had updated the mark 82 to an old student name “Deepika” .
- The above mentioned screenshot represents the result.

Task 5: Use the `keys()`, `values()`, and `items()` methods to print all keys, values, and key-value pairs in the `student_marks` dictionary.

Solution :

Preview screenshot :



The screenshot shows a Google Colab notebook titled "Python Assignment 2: Data Structures - List, Dictionary, Set & Conditional Statements.ipynb". The notebook contains three code cells, each with a comment and a corresponding Python command:

- Cell 1: Comment: "Use the keys(), values(), and items() methods to print all keys, values, and key-value pairs in the student_marks dictionary." Code: `print(student_marks.keys())`
- Cell 2: Code: `dict_keys(['Kavi', 'Savitha', 'Meena', 'Deepika', 'Suguna', 'Janani'])`
- Cell 3: Code: `print(student_marks.values())`
- Cell 4: Code: `dict_values([82, 91, 56, 82, 73, 80])`
- Cell 5: Code: `print(student_marks.items())`
- Cell 6: Code: `dict_items([('Kavi', 82), ('Savitha', 91), ('Meena', 56), ('Deepika', 82), ('Suguna', 73), ('Janani', 80)])`

The notebook interface includes a menu bar (File, Edit, View, Insert, Runtime, Tools, Help), a toolbar (Commands, Code, Text, Run all), and a status bar (Variables, Terminal, 4:20 PM, Python 3). The bottom of the image shows a Windows taskbar with various application icons and system information (29°C, Partly sunny, 16:20, 02-10-2026).

Explanation :

- I had used the `key` method to print only keys in the student mark dictionary .
- And I had used the `values` method to print only values in the student mark dictionary.
- And then I used the `item` method to print key- value pairs in the student mark dictionary.
- The above mentioned screenshot represents the result.

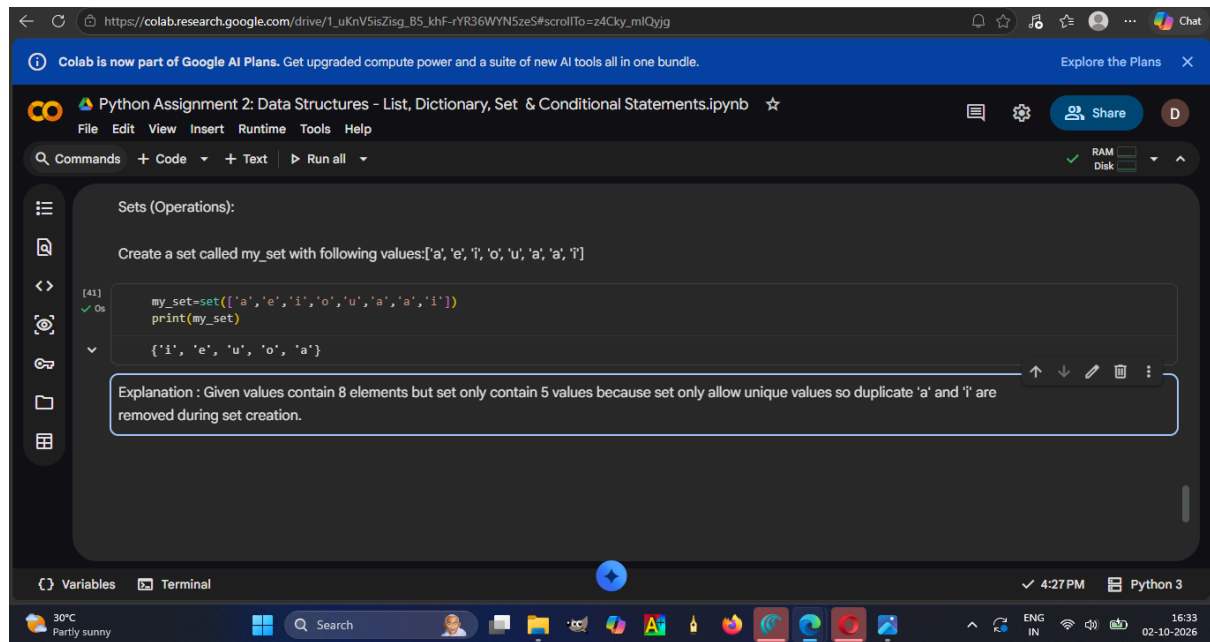
Sets (Operations):

Task 1 : Create a set called `my_set` with following values:

`['a','e','i','o','u','a','a','i']` . Analyse the output and provide explanation for the same.

Solution :

Preview screenshot :



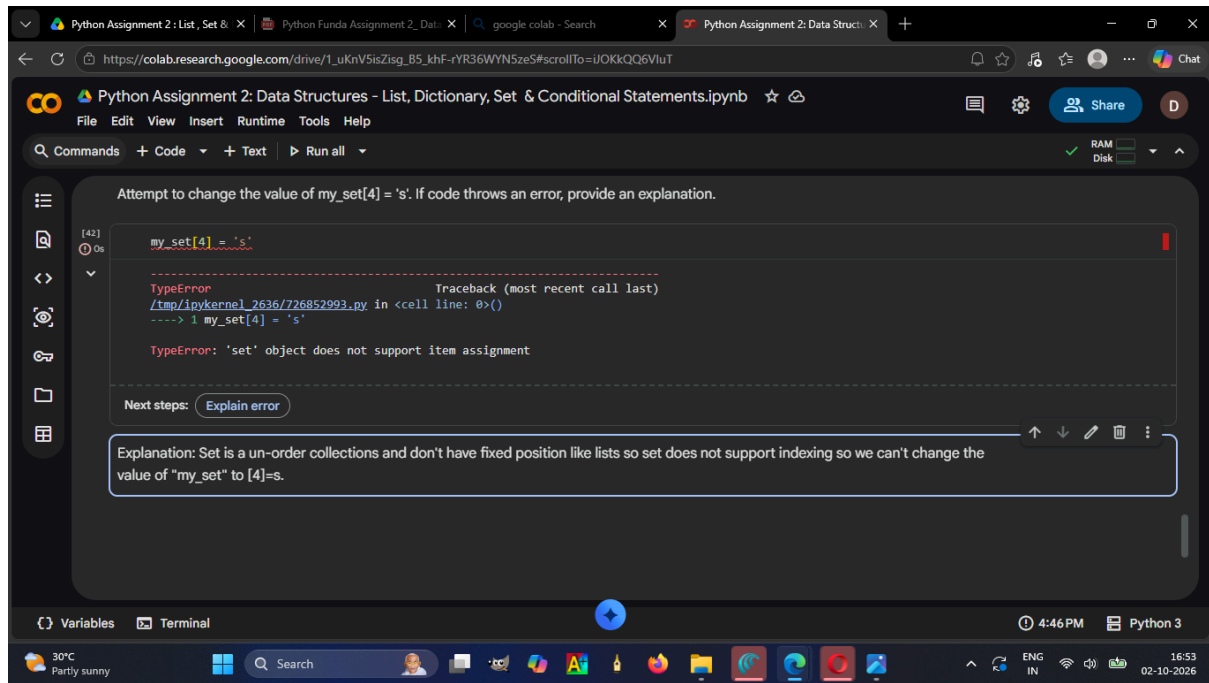
Explanation :

- I had created the set with a given value .
- Given values contain 8 elements but created sets only contain 5 values because sets only allow unique values so duplicates 'a' and 'i' are removed during set creation. .
- The above mentioned screenshot represents the result.

Task 2 : Attempt to change the value of `my_set[4] = 's'`. If code throws an error, provide an explanation.

Solution :

Preview screenshot :



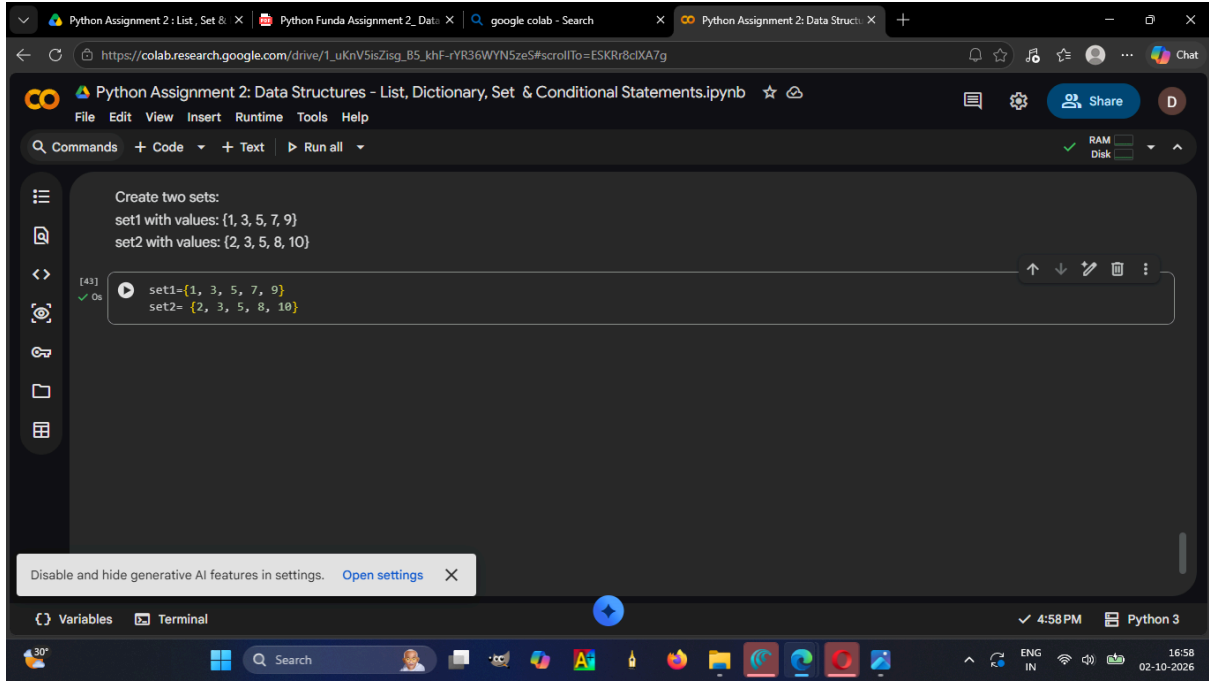
Explanation :

- I had attempted to change the value of `my_set[4]` but it showed an error because set is an un-order collection and doesn't have fixed position like lists so the set does not support indexing so we can't change the value of "my_set" to `[4]=s`.
- The above mentioned screenshot represents the result.

Task 3 : Create two sets: set1 with values: {1, 3, 5, 7, 9} , set2 with values: {2, 3, 5, 8, 10}

Solution:

Preview screenshot :



The screenshot shows a Google Colab notebook titled "Python Assignment 2: Data Structures - List, Dictionary, Set & Conditional Statements.ipynb". The notebook is open to a cell containing the following text:

```
Create two sets:  
set1 with values: {1, 3, 5, 7, 9}  
set2 with values: {2, 3, 5, 8, 10}
```

Below the text, there is a code cell with the following Python code:

```
set1={1, 3, 5, 7, 9}  
set2= {2, 3, 5, 8, 10}
```

The code cell is executed, and the output shows the two sets created. The notebook interface includes a menu bar (File, Edit, View, Insert, Runtime, Tools, Help), a toolbar (Commands, + Code, + Text, Run all), and a status bar (Variables, Terminal, 4:58 PM, Python 3). The bottom of the screen shows the Windows taskbar with various application icons.

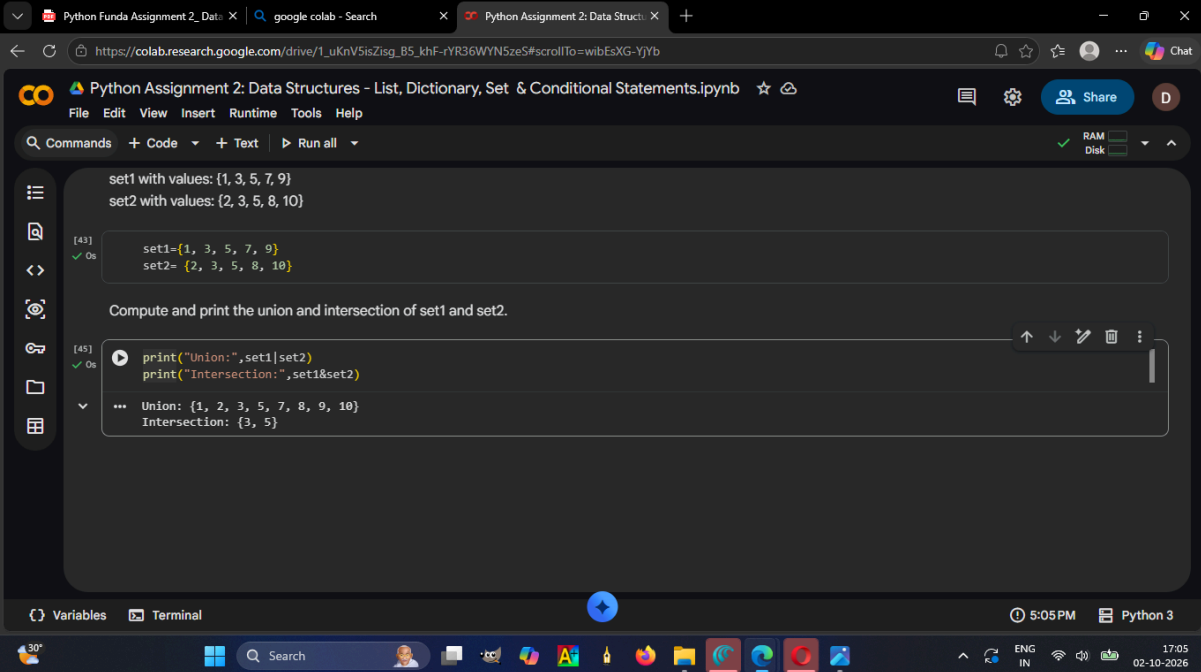
Explanation :

- I had created the two sets named set1 and set2 with the given values.
- The above mentioned screenshot represents the result.

Task 4 : Compute and print the union and intersection of set1 and set2.

Solution:

Preview screenshot :



The screenshot shows a Google Colab notebook titled "Python Assignment 2: Data Structures - List, Dictionary, Set & Conditional Statements.ipynb". The notebook contains the following code and output:

```
set1 with values: {1, 3, 5, 7, 9}
set2 with values: {2, 3, 5, 8, 10}
```

```
[43] ✓ Os
set1={1, 3, 5, 7, 9}
set2= {2, 3, 5, 8, 10}
```

Compute and print the union and intersection of set1 and set2.

```
[45] ✓ Os
print("Union:",set1|set2)
print("Intersection:",set1&set2)
```

```
... Union: {1, 2, 3, 5, 7, 8, 9, 10}
Intersection: {3, 5}
```

The bottom of the screenshot shows the Windows taskbar with the time 5:05 PM and date 02-10-2026.

Explanation :

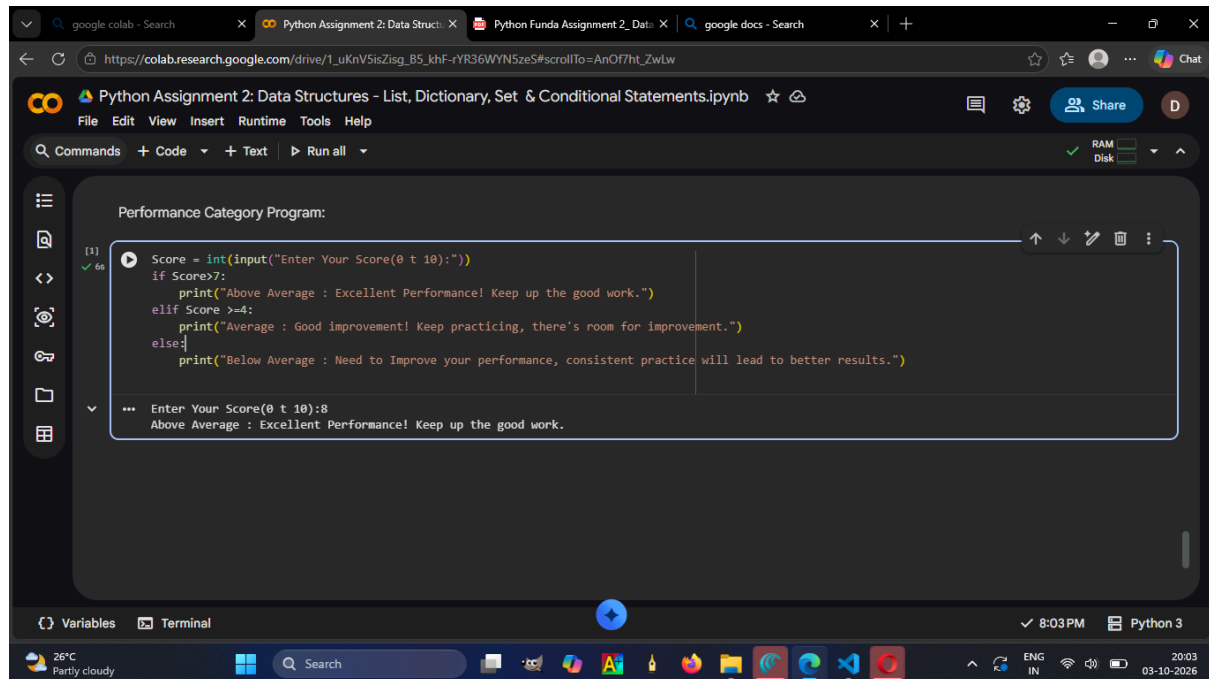
- I had used the “|” symbol for performing union and “&” symbol for performing intersection with set1 and set2 .
- The above mentioned screenshot represents the result.

Operators & Conditional Statements : (IF, ELIF, ELSE)

Task :Write Performance Category Program:

Solution :

Preview screenshot :



The screenshot shows a Google Colab notebook titled "Python Assignment 2: Data Structures - List, Dictionary, Set & Conditional Statements.ipynb". The code in the notebook is as follows:

```
Performance Category Program:

[1] Score = int(input("Enter Your Score(0 t 10):"))
    if Score>7:
        print("Above Average : Excellent Performance! Keep up the good work.")
    elif Score >=4:
        print("Average : Good improvement! Keep practicing, there's room for improvement.")
    else:
        print("Below Average : Need to Improve your performance, consistent practice will lead to better results.")

... Enter Your Score(0 t 10):8
Above Average : Excellent Performance! Keep up the good work.
```

The output of the program is "Above Average : Excellent Performance! Keep up the good work." The notebook interface includes a left sidebar with icons for file management, a top menu bar with options like File, Edit, View, Insert, Runtime, Tools, and Help, and a bottom status bar showing system information like temperature and time.

Explanation :

- I had created the nested conditional statement for the performance category program.
- First it will ask the input score with a message that "The score range should be from 0 to 10 : " .
- Then it will run the if and compare the input with a condition greater than 7 .
- If the score is greater then 7 the print the result as "Above Average : Excellent Performance! Keep up the good work."
- If the score is not greater than 7, and it is greater than or equal to 4 ,it prints the result as "Average : Good improvement! Keep practicing, there's room for improvement."
- If the score is less than 4 then it will print the result as "Below Average : Need to Improve your performance, consistent practice will lead to better results."
- The above mentioned screenshot represents the result.

